INDIVIDUAL WEEKLY PROJECT REPORTS

Student: Annalakshmi R

Role: Data Preprocessing and Exploratory Data Analysis (EDA)

Project Title: Flight Delay Analysis and Visualization

WEEK 4 – Handling Invalid Values and Outliers

During the fourth week, I concentrated on examining unusual values and potential outliers in the dataset. Outliers are observations that differ significantly from the general pattern of the data. In flight delay analysis, unusually large delays may occur due to genuine operational problems, while certain extreme values may also result from incorrect entries.

I examined numerical variables related to flight timings and delays to understand their distribution and identify unusual observations. I understood that simply removing every extreme value would not be appropriate because some unusually long delays could represent actual flight conditions.

I also explored how statistical methods and visualizations could help identify outliers. Techniques such as box plots and the interquartile range can be useful for examining the spread of numerical data and identifying observations that fall outside the expected range.

During this stage, I discussed the treatment of unusual values with the team and considered the importance of distinguishing genuine extreme observations from invalid records. I also understood that the decision to remove or retain an observation should depend on its meaning and the context of the dataset.

I documented the observations and ensured that the preprocessing process was consistent with the objectives of the project.

Outcome and Learning:

This week improved my understanding of outlier detection and data validation. I learned that unusual observations should be investigated carefully instead of being removed without understanding their significance.